Fiscal Year:	FY 2017	**Task Last Updated:**	FY 04/18/2017
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Project Title:	Cyclonic Two-Phase Flow Separator Experiment in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	06/01/2014	**End Date:**	05/30/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kadambi, Jaikrishnan Ph.D. (Case Western Reserve University)		
Grant/Contract No.:	NNX14AJ74G		
Performance Goal No.:			
Performance Goal Text:			
	Multiphase flow separation is a critical technological challenge for the human exploration of space and planetary bodies. While droplet and bubble segregation is a spontaneously occurring phenomenon in most terrestrial situations, the absence of the gravity-induced separatory action in a microgravity environment often results in situations where two disparate phases have no distinct inclination to separate from one another. This can present many problems with respect to spacecraft and space station operations as proper control and sequestration of the individual phases is often essential to the successful functionality of many system-critical components. Passive cyclonic separators are a promising technology that potentially could fulfill the separation needs of many of these systems. While considerable research has been performed on this technology, the limitations of terrestrial testing, aircraft testing, and modeling necessitates long-term experimentation to raise these devices to a sufficient technology readiness level (TRL) to permit use within spacecraft systems. It is a lack of basic engineering and scientific knowledge		

Task Description:

that currently prevents these systems from being employed. At a basic engineering level, there is a distinct lack of data regarding both the steady and unsteady operational performance of vortex separators.

Issues such as where the designs can be expected to work and how to go about optimizing designs are unresolved. Accurate prediction of these phenomena using computational fluid dynamics (CFD) is currently not tenable due to the dearth of available data and incomplete modeling techniques. Beyond engineering, there is also a lack of knowledge regarding the fundamental multiphase fluid physics phenomena that occur within passive cyclonic separators. Issues such as gas core adjustment to injection flow rates, bubble dynamics, vortex flow interaction with capillary forces at the gas core, etc. are all unresolved. Inasmuch, this research effort seeks to resolve these issues. The main objectives are to:

1. Quantitatively assess and delimit the range of operability of a prototype passive cyclonic separator including the transient and steady-state instability behavior of the device with respect to both device and system-level conduct.

2. Elucidate and quantify the multiphase fluids physics phenomena governing the performance of microgravity two-phase separators.

3. Validate and verify the multiphase CFD modeling approaches that have been developed to simulate the complicated two-phase transport processes associated with phase separation in microgravity.

4. Validate and verify a hydraulic system code for the simulation of two-phase flow in microgravity.

To achieve these objectives, combined experimental, numerical, and phenomenological modeling analyses are being performed to quantitatively describe the performance and response of systems containing separators. Specifically, the work entails the creation of multiphase flow system that is able to provide metered quantities of gas and liquid into a test separator. The separator and the associated flow loop had been operated under a variety of specially tailored conditions, including specific combinations of parameters to examine flow features of interest including the steady-state performance margins of the separator, the transient response of the separator, vapor core instability and misting, system startup/shutdown, and system level performance and instabilities. The products of this research have included the flight experiment (including two-phase flow loop and passive cyclonic separator), validated and verified CFD modeling techniques, a microgravity hydraulic system code, and archival data capturing the important fluid physics associated with microgravity separator operations.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

From a microgravity/scientific standpoint, the experimentation, modeling, and analytical work of this research will serve to:
1. Provide elucidation and understanding of the above-mentioned two-phase flow phenomena associated with the operation of separators in multiphase systems

2. Provide details of the columnar stability of the gas core

3. Provide details of the time constants associated with the response of passive cyclonic separators

4. Validate and verify a coupled CFD and hydraulic system code for simulations of two-phase flow in microgravity

The resulting insight into phase separation, distribution, and control will help to afford NASA designers the latitude to take greater advantage of the benefits offered by the use of these multiphase systems in spacecraft applications. The experimental results will also facilitate further new and original design modifications to improve the performance of microgravity phase separation devices. Validation and verification of the numerical results are expected to permit the use of CFD as a reliable design tool to further assist in the creation and analysis of new separator designs, while the results from the system code are expected to assist in the creation and analysis of general multiphase systems in microgravity.

Overall, the research will help contribute to the fundamental scientific understanding of two-phase fluid behavior, both in microgravity and 1-g environments.

Numerical Modeling
The computational studies of the fluid physics within the separation device conducted during the past year of the grant have been continued. A hybrid multiphase CFD solver was continuously improved and extended to simulate time-dependent 3D complex fluid flow in the separator. This hybrid multiphase model (e.g., Lagrangian-VOF model) coupled with Large Eddy Simulation was implemented utilizing the OpenFOAM libraries that enable multiphase CFD approach to model the dynamics of gas bubbles interacting with a turbulent rotating liquid film.

In this hybrid approach, the dispersed bubble is simulated using Lagrangian method while the gas core and annular liquid within are treated as Eulerian continua. The VOF (volume of fluid) model is applied to describe the fluid behavior of the gas core, annular liquid, and the interface between them. The Lagrangian bubble is connected to the Eulerian phases with a two-way coupling through interfacial force (or momentum exchange) term in the momentum equation. The effects of the fluid turbulence on the bubble motion are taken into account and modeled by an eddy interaction model.

Numerical simulations of the bubble motion and its residence time in the separator are carried out to provide more detailed information on the separation process. The effects of bubble size and inlet liquid flow rate on separator efficiency are examined. The simulation result demonstrates that turbulence plays an important role in the bubble motion. In this hybrid model, the number of bubbles which are collected by the gas core can be determined so that the separation efficiency of the separator can be quantified.

Analytical Modeling

The analytical approach has continued developing to provide insight into device design. In a microgravity environment, surface tension destabilizes the gas core, which adversely affects the performance. The control-volume-based analytical approach (initially developed in previous years) has been extended to account for capillary effects. The modifications required the addition of terms to account for pressure gradient induced by surface tension force at the interface.

Task Progress:	This surface tension effect is investigated analytically as well as numerically. Liquid-only injection is investigated herein to learn the basic physics of the surface tension effect. The collapse of the gas core due to surface tension is investigated using control volume analysis to determine the condition for the gas core collapse. No solution exists for the control volume equations below a certain Weber number, which is called the minimum Weber number for the stable gas core. The CFD analysis, based on the VOF scheme, shows stable as well as unstable gas cores. The gas core starts to collapse from the region near the bottom plate where the fluid velocity is reduced due to the no-slip condition. The CV predictions agree reasonably well with the CFD results so that the control volume analysis can be used to design the microgravity experiment. Experimentation Various ground-based experimentations have been continued to support to support the planned ISS experiment through the past year. The experimental works consist of mainly two tasks: (1) a continuation of characteristic bubble size distribution, and (2) the characteristic of the gas-liquid free interface motion. Those works are briefly described below. The experimental study on bubble size distributions is continued. A photographic technique with the high-speed camera is used to characterize bubbles size distribution at the separator inlet. Effect of the flow rate, volumetric quality, upstream two-phase flow structure, and upstream pipe length on the bubble size distributions are carried out. For the study the motion of gas-liquid interface in the separator, the experimental technique has been developed to fully characterize the dynamics of the interface with the focus on interface wavelength, frequency, and estimates of amplitudes. Supporting Work for Flight Hardware Design A breadboard model of the proposed experiment is being designed and built at NASA Glenn Research Center (GRC). We have collaborated with NASA/GRC and ZIN engineers to prepare and design the breadboard experiment system in past year. Planned Work and Facility Utilization Extensive collaboration with NASA personnel and utilization of NASA facilities is to continue in the upcoming year. Preparation for the eventual flight experiment will be continued in the coming year in concert with the above studies. The experimental study on interface motion is ongoing. The computational work is also to continue at Case Western Reserve University (CWRU) in the upcoming year with the objective of augmenting the maturity of CFD as a design tool for this class of fluid flows.
Bibliography Type:	Description: (Last Updated: 03/06/2024)
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Dissertations and Theses	Kang M-F (Ming-Fang Kang) "Investigation of Passive Cyclonic Gas-Liquid Separator Performance for Microgravity Applications." Ph.D. Thesis, Case Western Reserve University, Cleveland, Ohio, January 2017. , Jan-2017